

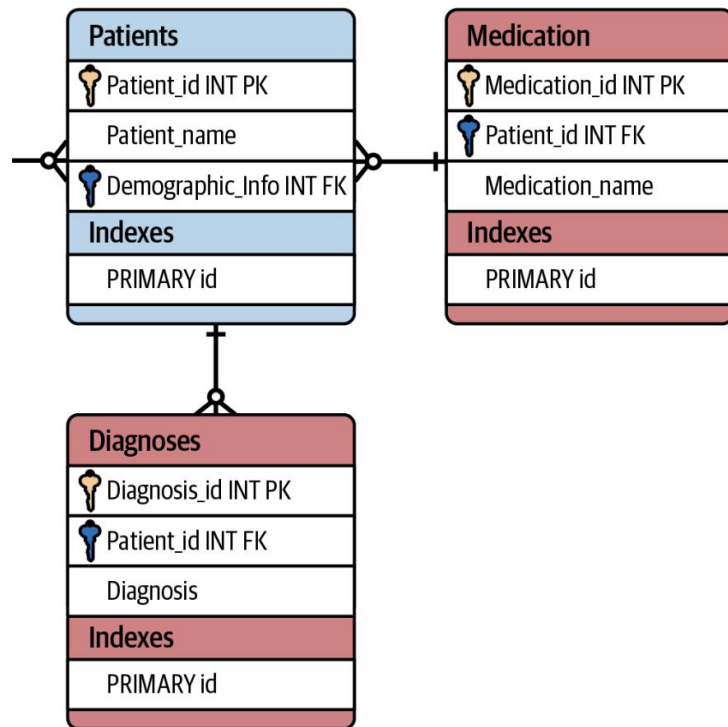
WINTER SEMESTER 2026

Foundations of Biomedical Informatics

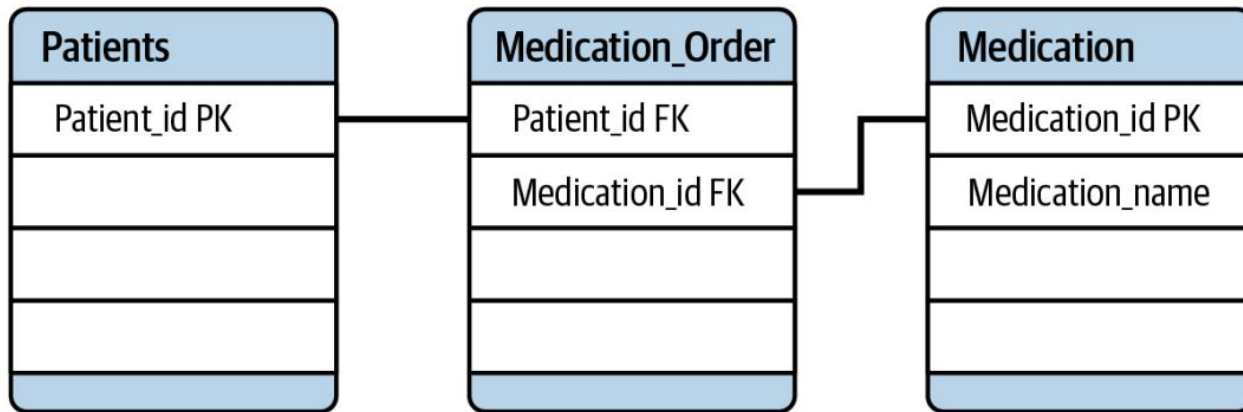
Lecture 5: Data Standards



Learning Health Systems and Databases (e.g. MIMIC)



What will you do with Such Databases?



Assignment Q1:

Pull out All Ways in which Insulin is Present in MIMIC IV

The PRESCRIPTIONS table contains medication orders that are entered within the hospital's order entry system (also known as the computerized provider order entry, or CPOE, system). Of course, there are typical columns you would expect, such as time- stamps for starting or discontinuing the medication, or the ICU stay or hospital admission with which the order is associated.

ndc	drug
63323026201	Heparin
00641040025	Heparin
17191003500	Heparin CRRT
17191003500	Heparin Flush
08290036005	Heparin Flush (10 units/mL)
00409115170	Heparin Flush (10 units/mL)
64253033335	Heparin Flush (100 units/mL)
63323026201	Heparin Flush (5000 Units/mL)
00641040025	Heparin Flush CRRT (5000 Units/mL)
63323026201	Heparin Flush CRRT (5000 Units/mL)
64253033335	Heparin Flush CVL (100 units/mL)
64253033335	Heparin Flush Hickman (100 units/mL)
64253033335	Heparin Flush PICC (100 units/mL)
00409115170	Heparin Lock Flush
00338055002	Heparin Sodium
00074779362	Heparin Sodium
00264958720	Heparin Sodium
00409779362	Heparin Sodium
63323054207	Heparin Sodium

Beginning at the Beginning

PAST EYE HISTORY:

X

GEN. MEDICAL HISTORY (F.H.)

Edible & hypotension

ALLERGIES:

Sulf

OCULAR EXAMINATION:

NW -2.00 DS +1.35 add

VISUAL ACUITY:

-1.50 DS +1.35 add

REFRACTION:

Present
(glasses)

SC 20/40
40-2

EC 2
100

Manifest

R1A -2.75 DS → 20/40 +1
-3.00 DS → 20/40 +1

Cycloplegic

STANFORD UNIVERSITY HOSPITAL
STANFORD UNIVERSITY MEDICAL CENTER
STANFORD, CALIFORNIA 94305

CLINIC HISTORY

(addressograph stamp)

Present illness: (date) June 3, 1989 Chief Complaint:

Admission Note

ID: 1st admission for this 22 y/o Mexican American F who presents with

CC: headache for one week

HPI: On 5/29 pt noted the onset of myalgias, severe headache, nausea, neck pain, and shaking chills. She consulted her private MD for these problems, and he diagnosed migraines & prescribed a combination med (villadonna, askaleids), phenobarbital, and ergotamine tartrate plus myprokamate. However, her sx worsened over the next week until 6/3 when she presented to our ER. She denies photophobia, diplopia, & other neurologic symptoms. She has noted a nonproductive cough but is a nonsmoker and she denies hemoptysis. She denies exposure to diseased individuals, specifically including meningococcal disease or TB.

PMH: No hx of illnesses other than NCD's. Meds only as above. Allergies: ⊕ Surgery ⊕ One daughter, age 12, by NVD.

Social: Married 14 yrs. Works in home. Has never lived in San Joaquin Valley. Last travelled to Mexico by car in 1974.

RDS: Genl: well until 10 days PTA

Skin: ⊕

Head: ⊕ F for HPT.

18-899
(Rev. 3/84)

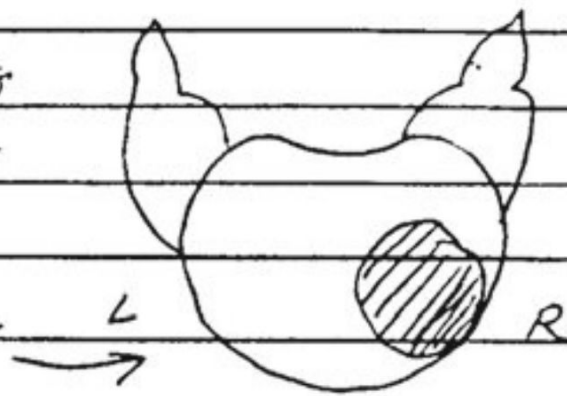
(Signature)

M.D.

TESTES - N/ size and consistency; no masses

RECTAL - 0 masses; (R)

prostate nodule
~ 1.5 cm



“The chart is not the patient.”

Gall, 1986



Enabling Information and Communication is Key



Standardization and Health Care

AUG 18 1972

J. H. U. BROWN, SENIOR MEMBER, IEEE, AND DEWITT JAMES LOWELL

NON-CIRCULATING

Do Not Remove
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Abstract—In order to deliver reasonable health care to all people, it is essential that standards be established. Standards vary with the type of control and with the approach desired in determining the quality of care. This paper discusses various kinds of standards and their application in the health care field. Standards may be determined as a process or as a direct regulation. It is probable that regulation of standards by process is the most satisfactory method.

arbiter may be the market place or agencies that rely on expertise from many sources to set acceptable standards of quality or performance. For these reasons, the final moderator may be found in a governmental authority, and its delegation into a system of regulation, law, and judicial action, so that an established code can become the focal point of resolution.

INTRODUCTION

SOCIETY cannot exist without a yardstick by which its accomplishments or failures are measured. Such yardsticks are called *standards*. They are created by the need for regulation and control as an escape from anarchy or to motivate towards greater achievement. In the ultimate, society dictates these limits by the demands it places upon itself. Standards provide opportunities for security and augmentation of process and output by virtue of the goal and process structure that they provide.

THE OBJECTIVES OF STANDARDIZATION

Standards have value within themselves in that they help establish quality. However, they accomplish more for society than the mere establishment of a level of quality and performance. A standard allows coordination of effort between producers so that like products can be produced. It permits the reproduction of similar units in mass quantity and permits the consumer to judge one product or service against another by performance. It establishes *freedom of interchange* of material and ideas, and permits the activity in one part of society

What do Standards Help With?

Healthcare involves many actors, across time, roles, and locations

- **Enables accurate communication between actors, roles, locations**
- **Standards ensure shared meaning and interpretation**
- **Comparability of data across systems and institutions**
- **Interoperability between health information systems**
- **Reusability of data for care, research, and analytics**
- **Improved efficiency of healthcare services**
- **Reduced errors by avoiding duplicated data entry**